Continued…
The Threat Environment: Attackers and Their Attacks

Reference:

Corporate & Network Security, Chapter 1

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Agenda

- The threat environment—attackers and their attacks
- Basic security terminology
- Employee and ex-employee threats
- Traditional external attackers
- The criminal era and competitor threats
- Cyberwar and cyberterror

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Employee and Ex-employee threats

- Are “insiders” the biggest threat ?

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- Threat can be from two sides
- Internal or External Attacks.
- Let have an example to protect home.
- Guard , wires are there to protect home.
- Guard knows each and every thing.

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2.2Employee and Ex-Employee Threats

- **2.2.1Employees and Ex-Employees Are Dangerous**
  - Dangerous because
    - They have knowledge of internal systems
    - They often have the permissions to access systems
    - They often know how to avoid detection
    - Employees generally are trusted
  - IT and especially IT security professionals are the greatest employee threats
    - "Who will guard the guards themselves?"

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Employee and Ex-Employee Threats

Employees and ex-employees are dangerous because they have extensive knowledge of systems, have the credentials needed to access sensitive parts of the systems, often know how to avoid detection, and can benefit from the trust that usually is accorded to "our people"

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Employee and Ex-Employee Threats

Sophisticated computer knowledge not required

In 23 financial cybercrimes committed between 1996 and 2002, 87 percent were accomplished without any “sophisticated programming” – Keeney

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Employee and Ex-Employee Threats

The US department of justice has a website http://www.cybercrime.gov

which lists federal cybercrime prosecutions. Roughly half the defendants are IT or IT security employees or ex-employees

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Employee and Ex-Employee Threats

- **2.2.2 Employee Sabotage**
  - Destruction of hardware, software, or data
  - Plant time bomb or logic bomb on computer

Sabotage comes from the French word for "shoe" because disgruntled workers in the early years of the Industrial Revolution supposedly threw their wooden shoes into machines to stop production

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Employee Sabotage Example…

Tim Lloyd, a computer systems administrator, was fired for being threatening and disruptive. In retaliation, Lloyd planted a logic bomb on a critical server. When pre-set conditions occurred, the logic bomb destroyed the programs that ran the company's manufacturing machines. Lloyd also took home and erased the firm's backup tapes to prevent recovery..

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Lloyd's sabotage resulted in USD 10 million in immediate business losses, USD 2 million in reprogramming costs, and 80 layoffs. The attack led to a permanent loss of the company's competitive status in the hi-tech instruments and measurements market because the company could not re-build the proprietary software it had been using.

-Sharon Gaudin, Computerworld, 2002

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Another Sabotage Example…

Two traffic Engineers working for the city of Los Angeles pleaded guilty to hacking the city's traffic center and disconnecting traffic signals at four LA's busiest intersections. They then locked out the controls to these intersections so that it took four days to restore control.

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…Another Sabotage Example

They did this a few hours before their union's scheduled job action against the city in support of contract negotiations. For this infraction, they received 240 days of community service, and were required to have their computers at home and work monitored.

-Dan Goodin, 2008

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Employee and Ex-Employee Threats

- **2.2.3 Employee Hacking**
  - Hacking is intentionally accessing a computer resource without authorization or in excess of authorization
  - Authorization is the key

First documented use of the word “hacker” was in Steve Levy’s book, Hackers in 1984…

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Employee and Ex-Employee Threats

- Note that the motivation for hacking is irrelevant
- Penalties are the same whether you were trying to steal a million dollars or were merely "testing security"

Access has to be intentional, damage does not…

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Employee and Ex-Employee Threats

- **2.2.4 Employee Financial Theft**
  - Misappropriation of assets
    - E.g. assigning them via computer to themselves
  - Theft of money
    - E.g. manipulation of an application to be paid a bonus

Two accountants at Cisco Systems, illegally accessed a corporate computer to issue themselves USD 8 million worth of Cisco stocks.
-http://www.cybercrime.gov , 2001

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Employee and Ex-Employee Threats

- **Employee Financial Theft**

In another case, Quitugua Sabathia, 31, of Vallejo, California, used her computer to embezzle more than USD 875,000 from the North Bay Health Care Group. A former accounts payable clerk at North Bay, she accessed the firm's accounting software and issued approximately 127 checks payable to herself and others. To conceal the fraud, she altered the electronic check register to make it appear that the checks had been payable to north bay's vendors.

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Employee and Ex-Employee Threats

- **2.2.5 Employee Theft of Intellectual Property (IP)**
  - Copyrights and patents (formally protected)
  - Trade secrets: plans, product formulations, business processes, and other info that a company wishes to keep secret from competitors

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Employee and Ex-Employee Threats

- **Employee Theft of Intellectual Property (IP)**

Intellectual Property (IP) is the information owned by the company and protected by law.

Trade secrets are pieces of sensitive information that a firm acts to keep secret from competitors; e.g. plans, price lists and customer lists, etc.

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Employee and Ex-Employee Threats

- **Employee Theft of Intellectual Property (IP)**

A former DuPont research scientist admitted downloading trade secrets worth USD 400 million. Only when he announced his decision to leave was his downloading behavior analyzed. The analysis found that he had downloaded 16,700 documents – 15 times more than the second-highest downloader. Most of these documents had nothing to do with his primary research area.

– PC World, 2007

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Employee and Ex-Employee Threats

- **2.2.6 Employee Extortion**
  - Perpetrator tries to obtain money or other goods by threatening to take actions that would be against the victim's interest

For example the employee might deploy a logic bomb on the company's computer.

Stealing Intellectual Property (IP) and demanding money for not passing on the information is also extortion.

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Employee and Ex-Employee Threats

- **Harassment of Other Employees**
  - Via e-mail
  - Displaying inappropriate material

Washington Leung left a firm and later logged into his ex-firm's servers using passwords given to him while employed there. He deleted over 900 files related to employee compensation. To frame a female co-worker, he gave her a USD 40,000 annual raise, and a USD 100,000 bonus. He created a hotmail account in the name of the female employee and sent senior managers an email containing information from the deleted files.
http://www.cybercrimes.gov

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Employee and Ex-Employee Threats

- **2.2.7 Other Types Of Abuse**
- **Internet Abuse**
  - Downloading inappropriate material, which can lead to harassment lawsuits and viruses
  - Downloading pirated software, music, and video, which can lead to copyright violation penalties
  - Excessive personal use of the Internet at work

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Employee and Ex-Employee Threats

- **Non-Internet Computer Abuse**
  - Unauthorized access to private personal data on internal systems by curious employees

This type of behaviour was detected in the 2008 Presidential election campaign and in several celebrity hospitalizations
-Los Angeles Times, 2008

A survey of 300 senior IT administrators in a London security conference and trade show found that one in three admitted to looking at confidential or personal information in ways unrelated to their jobs
-Computerworld 2008

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Employee and Ex-Employee Threats

- **Carelessness**
  - Loss of computers or data media containing sensitive information
  - Carelessness leading to the theft of such information

A Ponemon survey in 2008 found that 630,000 laptops are lost at airports each year. Although only some of these are corporate computers, airports are not the only place where laptops are lost, and lost media (US drives) can be just as damaging.

–Ponemon Institute

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Employee and Ex-Employee Threats

- **Other "Internal" Attackers**
  - Contract workers who work for the firm for brief periods of time
  - Workers in contracting companies

Contract workers often get credentials that are not deleted after their engagement ends.

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Employee and Ex-Employee Threats

- **Example Of Contract Workers**

Claude Carpenter, a 19 year old employee of a firm managing servers for the US Internal Revenue Service (IRS) planted a logic bomb on the servers after he learned he was about to be fired. The IRS would have been the real victim had his logic bomb succeeded. He also planted the code on his supervisors computer to frame the supervisor.

http://www.cybercrime.gov

(2001)

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